Proceedings of Machine Learning Research vol vvv:1–4, 2026

Can PAC-Bayes Explain Robotic Behavior Cloning?

Herbie Wright HERBIEW@ENGINEERING.UPENN.EDU
GRASP, University of Pennsylvania, Philadelphia, PA

Abstract

Behavior cloning has become a powerful approach for solving complex robotic tasks. Modern state-of-the-art policies use large neural networks trained via imitation learning on collected robot demos. However, practice has far outpaced theory. Current generalization theories fail to explain why large neural networks trained on a handful of independent demos generalize well, predicting extremely vacuous bounds. On the other hand, current theoretical analysis of regret bounds typically only explain why policies succeed once their expert-matching error is already low or do not apply to the robotic setting of deterministic experts in continuous action-spaces. This proposed project aims to bridge the gap and explain why robotic behavior cloning works from training to deployment. The proposed project is to use PAC-Bayes analysis and machinery to explain why certain policies generalize and others do not. Firstly, I propose a fast rate $\tilde{O}(\frac{\mathrm{KL}+\ln(\ln(NT)/\delta)}{NT})$ PAC-Bayes bound over policies with negligible empirical risk under standard behavior cloning assumptions (e.g. stability). This bound will also be extended to the common setting of learning from certain non-Markovian expert policies. Secondly, I propose a simple heuristic PAC-Bayes prior/posterior construction that can be applied to a trained diffusion policy. Thirdly, I propose experimentally verifying whether this analysis can explain any observable behavior cloning phenomena using Maniskill.

Keywords: PAC-Bayes, Generalization theory, Behavior Cloning, Imitation Learning

1. Related Works

PAC-Bayes The first PAC-Bayes bounds were proposed by McAllester (1998), and have since been used to construct non-vacuous generalization bounds for large neural networks (Dziugaite and Roy, 2017; Lotfi et al., 2022). However, these bounds are extremely uninformative for modern behavior cloning, which generalizes with as low as $N \approx 100$ trajectories. Modern PAC-Bayes theory has been extended to fast rate bounds (Catoni, 2007; Tolstikhin and Seldin, 2013) as well as bounds for temporally-correlated data (Eringis et al., 2024; Alquier et al., 2013; Zitouni et al., 2026). A $\sqrt{N}$ rate has been developed for imitation learning (IL) (Ren et al., 2021). However, no work has derived a fast NT-rate bound for IL. I believe Boucheron et al. (2009) would be useful.

Unlike other generalization bounds, PAC-Bayes requires defining *prior* and *posterior* distributions. Both weight-space Gaussians and compression priors are common prior choices. Data-dependent priors also exist (Dziugaite et al., 2021). Other work draws connections between better posterior distributions and the Hessian of the loss (Wang et al., 2020; Pitas and Arbel, 2022).

Theoretical Analysis of Modern Robotic Behavior Cloning Modern robotic behavior cloning aims to imitate an expert policy in a continuous action space. This can lead to unfavorable *compounding errors* (Simchowitz et al., 2025; Ross and Bagnell, 2010). While compounding errors can be circumvented in certain settings (Foster et al., 2024), exponentially-compounding error bounds are provably tight. Some work has found that the practice of action chunking and expert noising in tandem with an exponential input-to-state stability (EISS) assumption result in more favorable regret bounds (Zhang et al., 2026). This more recent works suggests that for policies using modern practices, small expert matching error (population risk) leads to low regret. However, this does not explain how low population risk is achieved in the first place.

© 2026 H. Wright.

Empirical Analysis of Modern Robotic Behavior Cloning On the empirical side of things, it has been found that while the actual training or test loss does not correlate well with success rate, the expert-matching error does (Allshire et al., 2026). This suggests we can analyze population risk as a proxy for task success. Recent empirical work also argues that practices like action chunking help because the expert is non-markovian (Zeng et al., 2026; Lazzati et al., 2026), a setting largely ignored in theoretical analyses.

2. Problem Formulation

Consider a data-generating distribution of trajectories from a closed loop expert policy:

$$U_t = \pi^*(X_t; \xi_t), \qquad X_{t+1} = f(X_t, U_t), \qquad X_0 \sim p(X_0), \qquad t = 0, ..., T-1, \tag{1}$$

where $\xi_t \in [0,1]^{n_\xi}$ are independent bounded random innovations, $U_t \in [0,1]^{n_u}$ are bounded actions, and $\text{supp}(p(X_0)) \subseteq [0,1]^{n_x}$ is a bounded initialization space. We can draw N draws of length T trajectories from this distribution to be our dataset. Then, consider imitation learning over a set of hypotheses $\pi \in \Pi$, where we want to match the expert actions according to some loss $l(u, u')$.

Assumptions. These assumptions are used for the PAC-Bayes bound over expert-matching risk:

1. All $\pi \in \Pi$ are L-Lipschitz and π^* is L-Lipschitz with respect to both x and ξ.
2. The dynamics are EISS with respect to both the actions u and the innovations ξ.
3. The loss l is continuous over $[0,1]^{n_u}$ and 1-Lipschitz (e.g. $l = \|u - u'\|$, or $l = \frac{1}{2}\|u - u'\|^2$).

A key component of PAC-Bayes bounds is the prior and posterior construction. I propose using Gaussians in neural network weight-space to describe both the prior and posterior. The Hessian of the empirical loss may be a useful quantity here.

3. Goals

Goal 1: PAC-Bayes bound The first goal is to construct a PAC-Bayes bound of the form:

$$R(Q) - \hat{R}(Q) \leq C_1 \sqrt{\hat{R}(Q) \frac{(D_{\text{KL}}(Q\|P) + \ln \frac{\nu}{\delta})}{NT}} + C_2 \frac{D_{\text{KL}}(Q\|P) + \ln \frac{\nu}{\delta}}{NT}, \tag{2}$$

where C_1, C_2 depend on Lipschitz and stability constants, and $\nu \in \mathcal{O}(\ln(NT))$. Importantly, this bound uses an effective sample size of NT/C instead of just N. Note that this becomes "fast rate" when $\hat{R}(Q) \approx 0$. *This bound should be extended to a certain class of non-Markovian policies.*

Goal 2: prior and posterior construction The second goal is to come up with a simple heuristic method for constructing the required prior and posterior distributions for PAC-Bayes analysis. I want to investigate whether one can analyze the infinite limit of training dynamics on temporally correlated data can be used to construct an effective prior. I also want to investigate whether approximating the Hessian of the empirical risk is useful for posterior construction.

Goal 3: experimental validation The third goal is to investigate whether the implications from this theoretical analysis can be used to explain empirical phenomena using simple Maniskill tasks (Tao et al., 2025). Specifically, I want to investigate: *(a) Action chunking and Non-markovian policies; (b) Observation-space and architectural decisions (point clouds vs rgb vs state).* If time permits, cross-task transfer could also be investigated.

References

Arthur Allshire, Himanshu Gaurav Singh, Ritvik Singh, Adam Rashid, Hongsuk Choi, David McAllister, Justin Yu, Yiyuan Chen, Huang Huang, Pieter Abbeel, et al. Scalable behavior cloning with open data, training, and evaluation. *arXiv preprint arXiv:2606.27375*, 2026.

Pierre Alquier, Xiaoyin Li, and Olivier Wintenberger. Prediction of time series by statistical learning: general losses and fast rates. *Dependence Modeling*, 1:65–93, 2013. doi: 10.2478/demo-2013-0004. URL https://doi.org/10.2478/demo-2013-0004.

Stephane Boucheron, Gabor Lugosi, and Pascal Massart. On concentration of self-bounding functions. *Electronic Journal of Probability*, 14: 1884–1899, 2009. URL https://projecteuclid.org/journals/electronic-journal-of-probability/volume-14/issue-none/On-concentration-of-self-bounding-functions/10.1214/EJP.v14-690.full.

Olivier Catoni. Pac-bayesian supervised classification: the thermodynamics of statistical learning. *arXiv preprint arXiv:0712.0248*, 2007.

Gintare Karolina Dziugaite and Daniel M. Roy. Computing nonvacuous generalization bounds for deep (stochastic) neural networks with many more parameters than training data. In *Proceedings of the 33rd Annual Conference on Uncertainty in Artificial Intelligence (UAI)*, 2017.

Gintare Karolina Dziugaite, Kyle Hsu, Waseem Gharbieh, Gabriel Arpino, and Daniel Roy. On the role of data in pac-bayes bounds. In *International Conference on Artificial Intelligence and Statistics*, pages 604–612. PMLR, 2021.

Deividas Eringis, John Leth, Zheng-Hua Tan, Rafael Wisniewski, and Mihály Petreczky. Pac-bayes generalisation bounds for dynamical systems including stable rnns. In *Proceedings of the AAAI conference on artificial intelligence*, volume 38, pages 11901–11909, 2024.

Dylan J Foster, Adam Block, and Dipendra Misra. Is behavior cloning all you need? understanding horizon in imitation learning. *Advances in Neural Information Processing Systems*, 37:120602–120666, 2024.

Filippo Lazzati, Kyle Stachowicz, William Chen, Alberto Maria Metelli, Andrew Wagenmaker, and Sergey Levine. Why does action chunking improve behavioral cloning performance in robotic control? *arXiv preprint arXiv:2608.02547*, 2026.

Sanae Lotfi, Marc Finzi, Sanyam Kapoor, Andres Potapczynski, Micah Goldblum, and Andrew G Wilson. Pac-bayes compression bounds so tight that they can explain generalization. *Advances in Neural Information Processing Systems*, 35:31459–31473, 2022.

David A McAllester. Some pac-bayesian theorems. In *Proceedings of the eleventh annual conference on Computational learning theory*, pages 230–234, 1998.

Konstantinos Pitas and Julyan Arbel. Cold posteriors through pac-bayes. *arXiv preprint arXiv:2206.11173*, 2022.

Allen Ren, Sushant Veer, and Anirudha Majumdar. Generalization guarantees for imitation learning. In *Conference on Robot Learning*, pages 1426–1442. PMLR, 2021.

Stéphane Ross and Drew Bagnell. Efficient reductions for imitation learning. In *Proceedings of the thirteenth international conference on artificial intelligence and statistics*, pages 661–668. JMLR Workshop and Conference Proceedings, 2010.

Max Simchowitz, Daniel Pfrommer, and Ali Jadbabaie. The pitfalls of imitation learning when actions are continuous. In Nika Haghtalab and Ankur Moitra, editors, *Proceedings of Thirty Eighth Conference on Learning Theory*, volume 291 of *Proceedings of Machine Learning Research*, pages 5248–5351. PMLR, 30 Jun–04 Jul 2025. URL https://proceedings.mlr.press/v291/simchowitz25a.html.

Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse kai Chan, Yuan Gao, Xuanlin Li, Tongzhou Mu, Nan Xiao, Arnav Gurha, Viswesh Nagaswamy Rajesh, Yong Woo Choi, Yen-Ru Chen, Zhiao Huang, Roberto Calandra, Rui Chen, Shan Luo, and Hao Su. Maniskill3: Gpu parallelized robotics simulation and rendering for generalizable embodied ai. *Robotics: Science and Systems*, 2025.

Ilya O Tolstikhin and Yevgeny Seldin. Pac-bayes-empirical-bernstein inequality. *Advances in Neural Information Processing Systems*, 26, 2013.

Huan Wang, Nitish Shirish Keskar, Caiming Xiong, and Richard Socher. Assessing local generalization capability in deep models. In *International Conference on Artificial Intelligence and Statistics*, pages 2077–2087. PMLR, 2020.

Michael Zeng, Abhinav Agarwal, Ajay Bati, Brian Lee, Siddharth Ancha, and Russ Tedrake. Revisiting open-loop execution in robotics: Toward reactive, higher-performing policies. *arXiv preprint arXiv:2608.15938*, 2026.

Thomas T Zhang, Daniel Pfrommer, Chaoyi Pan, Nikolai Matni, and Max Simchowitz. Action chunking and data augmentation yield exponential improvements in behavior cloning for continuous spaces. In *International Conference on Learning Representations*, volume 2026, pages 126689–126742, 2026.

Abdelkrim Zitouni, Mehdi Hennequin, Juba Agoun, Ryan Horache, Nadia Kabachi, and Omar Rivasplata. PAC-Bayesian reinforcement learning trains generalizable policies. In Tong Zhang, Miroslav Dudik, Martin Jaggi, Alekh Agarwal, Sharon Li, Dale Schuurmans, Jerry Zhu, Felix Berkenkamp, Hanze Dong, and Alberto Bietti, editors, *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 168416–168443. PMLR, 06–11 Jul 2026. URL https://proceedings.mlr.press/v306/zitouni26a.html.